

Supplement to: Using functional information for binary classifications

Pablo Martínez-Camblor^{1*}

¹Departments of Anesthesiology and Biomedical Data Science,
Geisel School of Medicine at Dartmouth, NH, USA

This document contains additional results/information related to this reasearh. In the current Section 1, we include a new set of simulations in which we evaluate the behavior of four additional competitors on the same four scenarios (Table S1 and S2). Then, in Table S3, we provide the AUCs obtained by the 10 explored criteria (5 in the main manuscript, 4 in this supplement, and the proposed PBC) computed from a large Monte Carlo sample of size $(n_0, n_1) = (2500, 2500)$ in each of the 16 considered scenarios (associated sample error below 0.01). For the PBC and kNN procedures, following the training-testing approach, we run two independent samples with this same size, $(n_0, n_1) = (2500, 2500)$.

We also include a simulation scenario informed by the real problem presented in the Section 5 of the main manuscript. We considered two situations. In the first, we assume that the trajectories drawn from the underlying negative and positive populations are equal. In the second, we assume a model which provides similar curves to those obtained in the real data. In both cases, considered sample size was the original available in the Cardiotoxicity problem. The training-testing approach was considered for the PBC and kNN approaches using, approximately, 1/3 of the sample for training, and the remaining 2/3 for testing. Section 2 of this document reports the results obtained in these simulations. Finally, in Section 3, we specify some details regarding the implementation of our simulations. The full R code is provided in separate files and in https://github.com/pmcamblor/functional_ROC.

* *Correspondence to:* Pablo Martínez-Camblor. 7 Lebanon Street, Suite 309, Hinman Box 7261, Lebanon, NH 03751, USA. E-mail: Pablo.Martinez-Camblor@hitchcock.org

1 Monte Carlo simulations

1.1 Additional competitors

We evaluate the behavior of four additional competitors on the previous considered models. New competitors are a ranking method, RK, based on [1], a functional k-nearest neighbors procedure, kNN (1/3 of the sample for training, 2/3 for testing), and two procedures based on functional depth methods; particularly in the Fraiman and Muniz [2], D-FM, and Mode [3], D-DM.

In Model I-a (real AUC=0.5), all the procedures but kNN and PBC (both implementing a training-testing approach) showed some level of overfitting, overall for the smallest sample size. D-FM showed very small bias (around 0.51), and D-MD got an average AUC of 0.54 for $(n_0, n_1) = (50, 50)$, but reduced to 0.51 for $(n_0, n_1) = (200, 100)$. Then, PBC was the best procedure in Model I-b, with F-MD in the second position. They flipped in Model I-c with F-MD the best and PBC the second. In Model I-d, RK and PBC were similar, although the absence of outliers made RK the winner. In type II models (a, b, c, and d), F-MD and PBC reported almost perfect results although, again, the absence of outliers put F-MD in the first position. Behavior of kNN, in general, was affected by the balance between negative and positive sample sizes.

In the Model III-a RK, D-FM, D-MD and PBC reached very good results, with kNN showing poorer results changing with the balance between the number of negative and positive. RK and PBC tied in the first position (despite PBC had some outlier for the smallest sample sizes). In Model III-b, and III-d RK and PBC were, again, the overall winners. D-MD was the best procedure in Model III-c, with PBC and D-FM tied in the second position, RK got poorer results while the position of kNN oscillated with the sample size. Model IV-a (real AUC=0.5) shows the slightly overfitting of RK, remaining procedures behaved adequately. PBC performed the best overall, although D-MD is the winner in Model IV-b, and Model IV-d.

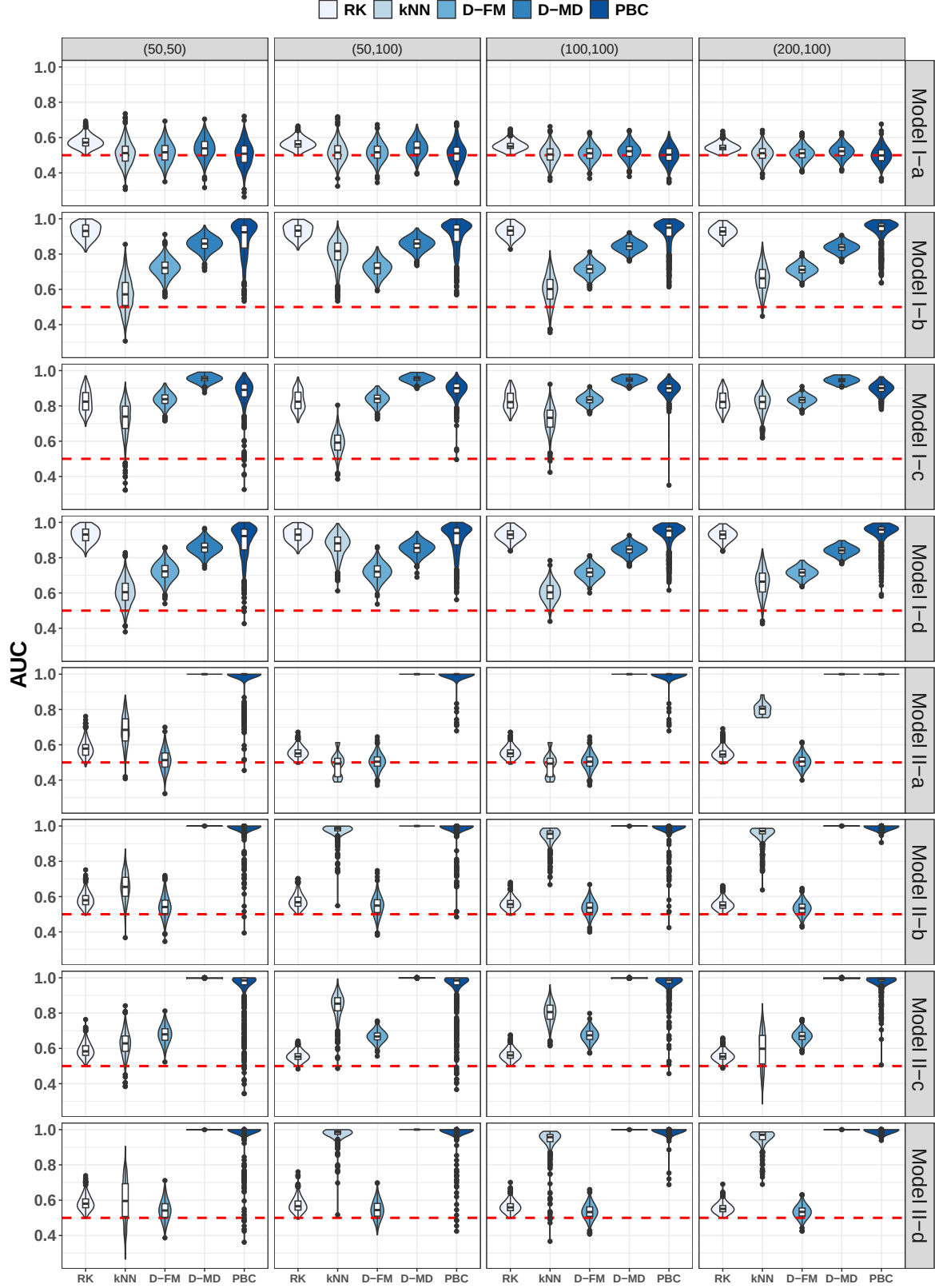


Figure S1. Models I-II. Violin plots for the AUCs observed in 1,000 Monte Carlo simulations for the models type I, and type II models, with different sample sizes.

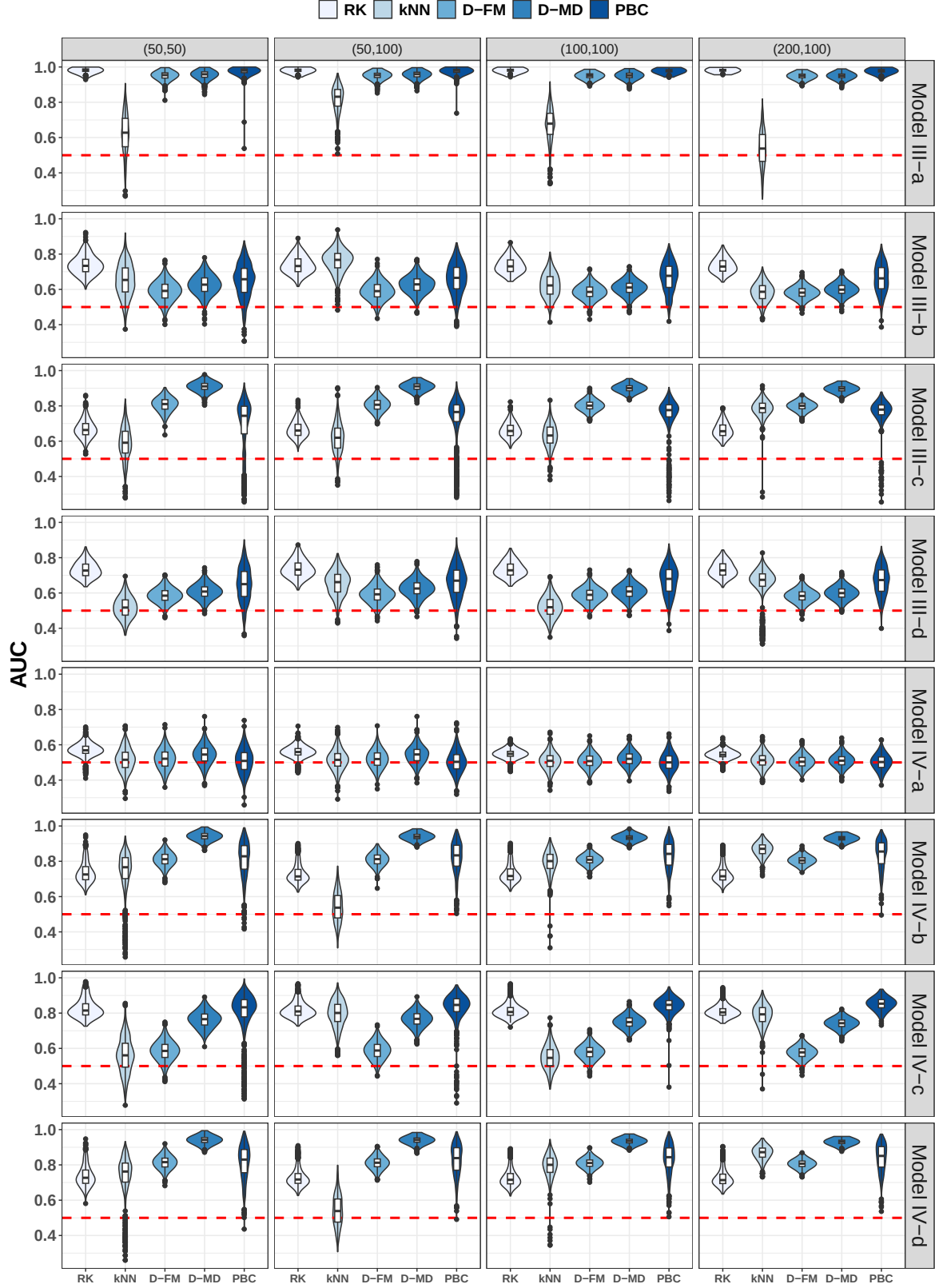


Figure S2. Models III-IV. Violin plots for the AUCs observed in 1,000 Monte Carlo simulations for the models type III, and type IV models, with different sample sizes.

1.2 Additional information

Table S1. AUCs obtained in the 16 models for the 10 criteria based on a large Monte-Carlo sample (size of $(n_0, n_1) = (2500, 2500)$). Notice that the AUC for the kNN procedure can strongly depend on n_1/n_0 .

	Min	Max	Int	Lin	Qua	RK	kNN	D-FM	D-MD	PBC
Model I										
-a*	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5
-b	0.88	0.74	0.49	0.97	0.98	0.93	0.68	0.71	0.83	0.96
-c	0.87	0.74	0.51	0.97	0.97	0.82	0.64	0.83	0.99	0.97
-d	0.88	0.74	0.49	0.97	0.98	0.93	0.75	0.72	0.90	0.96
Model II										
-a	0.76	0.75	0.51	0.51	0.50	0.53	0.65	0.51	1.00	1.00
-b	0.61	0.68	0.51	0.50	0.51	0.52	1.00	0.50	1.00	0.99
-c	0.70	0.71	0.50	0.51	0.75	0.58	0.75	0.67	1.00	0.99
-d	0.68	0.69	0.50	0.51	0.50	0.50	1.00	0.50	1.00	0.99
Model III										
-a	0.49	0.51	0.51	0.98	0.98	0.97	0.98	0.97	0.97	0.98
-b	0.45	0.49	0.50	0.75	0.76	0.79	0.63	0.60	0.62	0.76
-c	0.72	0.67	0.51	0.67	0.82	0.79	0.71	0.63	0.81	0.89
-d	0.44	0.49	0.49	0.75	0.76	0.78	0.55	0.58	0.59	0.77
Model IV										
-a*	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5
-b	0.42	0.74	0.53	0.68	0.82	0.71	0.81	0.80	0.95	0.86
-c	0.65	0.65	0.56	0.82	0.82	0.82	0.83	0.58	0.75	0.90
-d	0.43	0.75	0.55	0.70	0.82	0.72	0.55	0.83	0.97	0.87

* Real one, not based on a large Monte Carlo sample.

2 Real-problem supplements

First, we consider the situation (**Model-Null**) in which the functional marker has no relationship with the outcome. From the original 270 trajectories, and we replacement, we randomly select 243 and 27 functions for acting as the negative, and positive group, respectively. Then, we apply the 10 considered classification criteria Min, Max, Int, Lin, Qua, RK, kNN, D-FM, D-MD, and PBC. For the PBC and the kNN procedures, we implement the training-testing approach previously described using 80 negative, and 10 positive curves for constructing the model, and the remaining 163 and 17 curves for testing its accuracy. We replicate the process 1000 times. The observed coverage of the 95% CI percentage for the PBC procedure was 92.2%, with an average interval length of 0.274 ± 0.033 .

Then, we artifitially create an alternative simulation scenario, **Model-Alt**, which tries to emulate the curves observed in the real-world data. Figure S3 shows one random representation of this theoretical scenario with sample size $(n_0, n_1) = (243, 27)$ (the resulting curves were more or less similar to the real ones). Then, we generate random samples with the same original sample size. Again, we repite the process 1000 times. The coverage percentage of the 95% CI for the PBC procedure was 95.1%, with an average interval length of 0.174 ± 0.021 .

Figure S4 shows the violin plot for the AUCs in both the Model-Null and the Model-Alt. Results are consistent with the previous comments. In the Model-Null, Lin, Qua, RK and kNN show a relevant amount of overfitting, specially Qua and RK. In Model-Alt, Qua got good results and was the clear winner (average AUC of 0.88), followed by kNN (average AUC=0.80; potentially affected by the unbalanced sample size), and RK (average AUC=0.77). Then, D-FM and D-MD reached similar results, with an average AUC of 0.73. PBC got an average AUC of 0.68. Then, RK, Min and Max got very moderate results, with average AUCs around 0.62, slightly better than Lin. Finally, Int does not shown relationship with the outcome.

Figure S3. Example of positive (red), and negative (blue) random curves generated from the Model-Art. Sample size was $(n_0, n_1) = (243, 27)$.

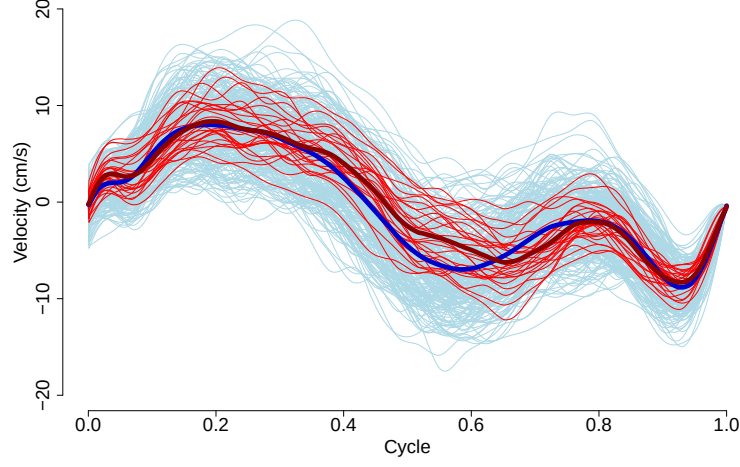
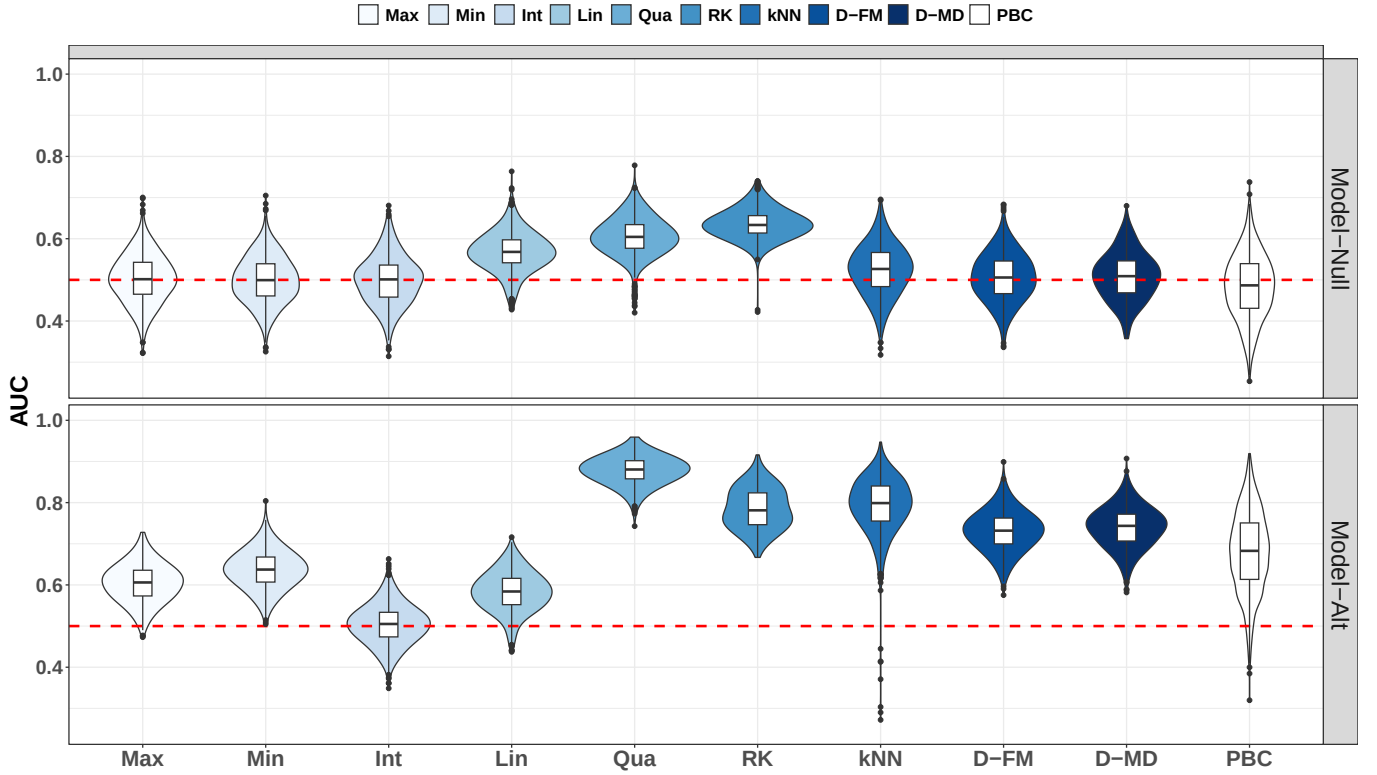


Figure S4. Violin plots for the scenarios Model-Null and Model-Alt; both informed by the real-world data example. Sample size was $(n_0, n_1) = (243, 27)$.



Further analyses of this data set, including informed Monte Carlo simulations, can be found in Martínez-Cambor and Díaz-Coto [4].

3 Computational details

The implementation process implied to take several decisions. The routine for implementing the Min, Max, and Int algorithms is very simple and straightforward. Routines implementing RK, kNN, D-FM, and D-DM were based on pre-existing codes and AI researches. No relevant comments to add.

We used the package `logitFD` for implementing Lin. and Qua. This package implements a logistic regression model for FD, and includes its own AUC compute. The process is based on a previous functional PCA analysis. We did this on the direct and on the quadratic transformed values of the raws:

```
>curves.fd <- Data2fd(argvals = t, y=t(DT),basisobj = FourierBasis)
>curves2.fd <- Data2fd(argvals = t, y=t(DT^2),basisobj = FourierBasis)
```

where `DT` contains the functions. Then, we include a number of components in the logistic regression. The final number was selected trying to avoiding very overfitted results and/or inconsistencies. Finally, we decided to select two components of each the linear, and the quadratic components:

```
lin <- logitFD.pc(Response= GR, FDObj=list(curves.fd), ncomp = 2)
quad <- logitFD.pc(Response= GR, FDObj=list(curves.fd, curves2.fd), ncomp = c(2,2))
```

For the PBC implementation, we used the packages `nsROC` [5] and looked for the best model between "both", "both2", and "left" in order to compute the 'individual' probabilities. This could be avoided when you deal with a problem in which you can use some rationality for selecting the adequate model. For instance, in the real-world example here considered, it was clear that the best model should be "both2" (both larger and lower distances could be associated with the negative group). For avoiding overfitting, we randomly select whichever criteria when the results provided were similar (difference below a threshold of 0.15).

References

- [1] G. Estévez-Pérez and P. Vieu. A new way for ranking functional data with applications in diagnostic test. *Computational Statistics*, 36(1):127–154, 2021.

- [2] R. Fraiman and G. Muniz. Trimmed means for functional data. *TEST*, 10(2):419–440, 2001. doi: 10.1007/BF02595706.
- [3] A. Cuevas, M. Febrero, and R. Fraiman. Robust estimation and classification for functional data via projection-based depth notions. *Computational Statistics*, 22(3):481–496, 2007. doi: 10.1007/s00180-007-0053-0.
- [4] P. Martínez-Camblor and S. Díaz-Coto. Exploring the use of functional data for binary classifications: The case of tissue doppler imaging in cardiotoxicity related-therapy cardiac dysfunction detection. *Axioms*, 15(2), 2026. doi: 10.3390/axioms15020120.
- [5] S. Pérez-Fernández, P. Martínez-Camblor, P. Filzmoser, and N. Corral. nsROC: An R package for non-standard ROC curve analysis. *The R Journal*, 10(2):55–77, 2018. doi: 10.32614/RJ-2018-043.